

# STAT452: Polynomial Regression & Transformation of Variables

Lesson 2 Notes

Nguyen Dinh Thien Loc

May 2026

## Contents

---

|          |                                                              |          |
|----------|--------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction to Polynomial Regression</b>                 | <b>2</b> |
| 1.1      | Motivation . . . . .                                         | 2        |
| 1.2      | Model Forms . . . . .                                        | 2        |
| <b>2</b> | <b>Important Considerations in Polynomial Regression</b>     | <b>2</b> |
| 2.1      | Order of the Polynomial Model . . . . .                      | 2        |
| 2.2      | Model-Building Strategy . . . . .                            | 2        |
| 2.3      | Extrapolation . . . . .                                      | 2        |
| 2.4      | Ill-Conditioning . . . . .                                   | 3        |
| 2.5      | Multicollinearity: Variance Inflation Factor (VIF) . . . . . | 3        |
| 2.6      | Hierarchy . . . . .                                          | 3        |
| 2.7      | Interpreting Coefficients in a Polynomial Model . . . . .    | 4        |
| <b>3</b> | <b>Example: The Hardwood Data</b>                            | <b>4</b> |
| <b>4</b> | <b>Transformation of Variables</b>                           | <b>4</b> |
| 4.1      | Why Transform? . . . . .                                     | 4        |
| 4.2      | Linear vs. Nonlinear Models (Revisited) . . . . .            | 4        |
| 4.3      | Linearisable Non-Linear Models . . . . .                     | 4        |
| 4.4      | Interpreting Log-Transformed Variables . . . . .             | 5        |
| 4.5      | Example: Bacteria Deaths Under X-Ray Radiation . . . . .     | 5        |
| 4.6      | Transformations to Reduce Skewness . . . . .                 | 5        |
| 4.6.1    | Power Transformation Family . . . . .                        | 5        |
| 4.6.2    | Rules of Thumb for Log Transformation . . . . .              | 5        |
| 4.6.3    | Effect of Transformations on Shape . . . . .                 | 6        |
| 4.7      | Diagnosing Non-Linearity: Always Check Residuals . . . . .   | 6        |
| 4.8      | Cautions When Transforming . . . . .                         | 6        |
| <b>5</b> | <b>Summary Table</b>                                         | <b>7</b> |

## Introduction to Polynomial Regression

---

### Motivation

The relationship between a response  $Y$  and a predictor  $X$  is often *not* a straight line. Polynomial regression is one of the simplest ways to model and test for departures from linearity.

The idea is to treat higher powers of  $X$  (e.g.  $X^2$ ,  $X^3$ ) as additional predictor variables. Despite the curved shape, the model remains **linear in its parameters**, so all standard MLR theory applies directly.

### Model Forms

**$k$ th-order polynomial in one variable:**

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k + \epsilon.$$

**Quadratic model in two variables (with interaction):**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \epsilon.$$

**Reformulation as MLR.** By substituting  $x_j = x^j$  (or  $x_j$  = an appropriate function of  $x$ ), every polynomial model can be rewritten in the standard MLR form  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$  and fitted using the Least Squares formula  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ .

**Matrix form (order-2 example,  $n$  observations):**

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

## Important Considerations in Polynomial Regression

---

### Order of the Polynomial Model

- A polynomial of order  $n-1$  can always fit  $n$  data points perfectly  $\Rightarrow$  this is almost certainly **overfitting**.
- **Principle of parsimony:** use the simplest model consistent with the data. Prefer a low-order model in a transformed variable over a high-order model in the original scale.
- Always try transformations first to keep the model first-order (linear) if possible.

### Model-Building Strategy

Two standard procedures:

- **Forward selection:** Start with the first-order model; add successively higher-order terms until the  $t$ -test for the highest-order term is non-significant.
- **Backward elimination:** Fit the highest-order model; remove terms one at a time (from highest to lowest order) until all remaining terms are significant.

**Note:** Forward selection and backward elimination do not necessarily yield the same final model. In practice, first- and second-order polynomials are most commonly used.

### Extrapolation

Polynomial models must be used with extreme caution outside the observed range of  $x$ . The curvature inside the data region can be very different from the curvature outside it, leading to wildly unreliable predictions.

### Ill-Conditioning

- In polynomial regression, the design matrix has multiple columns derived from the same predictor, creating **multicollinearity** (especially when  $x$  values are in a narrow range).
- As the polynomial order increases,  $\mathbf{X}^\top \mathbf{X}$  becomes increasingly **ill-conditioned**: matrix inversion is numerically unstable.
- **Remedy**: Centre the data by letting  $\tilde{x}_i = x_i - \bar{x}$  before forming higher-power terms. This removes “non-essential” ill-conditioning.

### Multicollinearity: Variance Inflation Factor (VIF)

The VIF for predictor  $X_j$  is:

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where  $R_j^2$  is the  $R^2$  from regressing  $X_j$  on all other predictors.

| VIF value | Interpretation                                          |
|-----------|---------------------------------------------------------|
| = 1       | No multicollinearity                                    |
| > 5       | Moderate multicollinearity                              |
| > 10      | Severe multicollinearity (coefficients may be unstable) |

### Hardwood data illustration.

| Model                        | Variable | $x$   | $x^2$ | $x^3 / x^4$   |
|------------------------------|----------|-------|-------|---------------|
| $y \sim x + x^2$             | VIF      | 1.097 | 1.097 | —             |
| $y \sim x + x^2 + x^3$       | VIF      | 6.13  | 1.29  | 6.82          |
| $y \sim x + x^2 + x^3 + x^4$ | VIF      | 7.80  | 14.71 | 11.49 / 19.58 |

Severe multicollinearity emerges for the fourth-order model.

### Recommendations to reduce multicollinearity:

- Use `poly(x, degree=n)` in R to create orthogonal polynomial terms.
- Standardise predictors before polynomial transformation.
- Consider regularised regression (Ridge/Lasso) for high-degree polynomials.
- Always monitor VIF.

### Hierarchy

A polynomial model is **hierarchical** if it contains *all* powers up to its highest order term. For example:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + \varepsilon \quad (\text{hierarchical})$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_4 x^4 + \varepsilon \quad (\text{NOT hierarchical; missing } x^3)$$

Only hierarchical models are invariant under linear transformation of  $x$ , so polynomial models should always satisfy this property.

## Interpreting Coefficients in a Polynomial Model

In a model that contains both  $x_1$  and  $x_1^2$ , it is **meaningless** to interpret  $\beta_1$  alone (it is impossible to change  $x_1$  while holding  $x_1^2$  fixed). Instead, interpret the *entire polynomial component* as a whole:

“The mean of  $y$  changes with  $x_1$  following the curve  $\beta_1 x_1 + \beta_2 x_1^2$ , holding all other predictors constant.”

## Example: The Hardwood Data

19 observations of kraft paper strength ( $y$ ) vs. percentage of hardwood ( $x$ ) in pulp. Four polynomial models were fitted and compared.

| Order         | $R^2$  | $R^2_{\text{adj}}$ | $\hat{\sigma}^2$ | $p$ -value (highest term) |
|---------------|--------|--------------------|------------------|---------------------------|
| 1 (linear)    | 0.3054 | 0.2645             | 11.822           | 0.0141                    |
| 2 (quadratic) | 0.9085 | 0.8971             | 4.422            | $1.89 \times 10^{-8}$     |
| 3 (cubic)     | 0.9707 | 0.9648             | 2.585            | $4.72 \times 10^{-5}$     |
| 4             | 0.9736 | 0.9661             | 2.539            | 0.233                     |

**Model selection reasoning:**

- The 4th-order term is not significant ( $p = 0.233$ ); VIF values are severe.
- The cubic model has highly significant highest-order term and modest VIF.
- **Selected model: cubic (3rd order)**, balancing fit and parsimony.

## Transformation of Variables

### Why Transform?

Variable transformations (of either  $Y$  or  $X$ ) address two core issues:

1. **Non-linearity:** the relationship between predictors and response is not linear. A suitable transformation can linearise it.
2. **Non-constant variance (heteroscedasticity):** the error variance  $\sigma^2$  is not constant across observations. Transformations can stabilise variance.

Common techniques: log, square root, reciprocal, Box-Cox family.

### Linear vs. Nonlinear Models (Revisited)

A model is *linear* if it is linear in its parameters, regardless of transformations applied to the predictors.

$$\begin{aligned} \textbf{Linear (in parameters)} \quad & Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon \\ & Y = \beta_0 + \beta_1 \log(X) + \varepsilon \\ & Y = \beta_0 + \beta_1 \sqrt{X} + \varepsilon \end{aligned}$$

$$\textbf{Nonlinear (in parameters)} \quad Y = \beta_0 + e^{\beta_1 X} + \varepsilon \quad (\text{cannot be transformed to linear})$$

### Linearisable Non-Linear Models

Many non-linear models can be made linear through appropriate transformations.

| Original model                                              | Transformation         | Linearised form                           |
|-------------------------------------------------------------|------------------------|-------------------------------------------|
| $Y = \alpha X^\beta$                                        | $\log Y, \log X$       | $\log Y = \log \alpha + \beta \log X$     |
| $Y = \alpha e^{\beta X}$                                    | $\log Y$               | $\log Y = \log \alpha + \beta X$          |
| $Y = \alpha + \beta \log X$                                 | $\log X$               | $Y = \alpha + \beta \log X$               |
| $Y = \frac{X}{\alpha X - \beta}$                            | $1/Y, 1/X$             | $1/Y = \alpha - \beta(1/X)$               |
| $Y = \frac{e^{\alpha + \beta X}}{1 + e^{\alpha + \beta X}}$ | $\log \frac{Y}{1 - Y}$ | $\log \frac{Y}{1 - Y} = \alpha + \beta X$ |

## Interpreting Log-Transformed Variables

**Semi-log model.**

$$\log(Y) = \beta_0 + \beta_1 X \Rightarrow Y = e^{\beta_0} e^{\beta_1 X}.$$

**Interpretation:** Each 1-unit increase in  $X$  *multiplies*  $Y$  by  $e^{\beta_1}$ .

**Log-log model.**

$$\log(Y) = \beta_0 + \beta_1 \log(X) \Rightarrow Y = e^{\beta_0} X^{\beta_1}.$$

- When  $X$  doubles,  $Y$  is multiplied by  $2^{\beta_1}$ .
- $\beta_1$  is the **elasticity**: a 1% increase in  $X$  leads to a  $\beta_1\%$  increase in  $Y$ .

## Example: Bacteria Deaths Under X-Ray Radiation

Biological theory suggests exponential decay:

$$N_t = N_0 e^{\beta_1 t}, \quad \text{taking } \log : \quad \log N_t = \underbrace{\log N_0}_{\beta_0} + \beta_1 t.$$

Fitted in R via `lm(log(N.t) ~ t, data = bact)`.

Key result:  $\hat{\beta}_1 = -0.2184$ , so the bacteria count decreases by approximately

$$1 - e^{-0.2184} \approx 19.6\%$$

every 6 minutes, with 95% CI approximately (18.5%, 20.8%).

## Transformations to Reduce Skewness

### 4.6.1 Power Transformation Family

$$f_\alpha(y) = \begin{cases} y^\alpha, & \alpha \neq 0, \\ \log(y), & \alpha = 0. \end{cases}$$

- **Right-skewed** distribution ( $\alpha < 1$ ):  $y \rightarrow 1/y, \log y, \sqrt{y}$ , or  $y^\alpha$  with  $\alpha < 1$ .
- **Left-skewed** distribution ( $\alpha > 1$ ):  $y \rightarrow y^2, y^3$ , or  $y^\alpha$  with  $\alpha > 1$ .

### 4.6.2 Rules of Thumb for Log Transformation

1. Use log when the variable measures an **amount of money** (income, prices, salaries).
2. Use log for variables measuring **concentration** or **rate** (e.g. chemical concentration in blood, population density).
3. Use log when a variable spans **several orders of magnitude** (e.g. from 10 to 10 000) to compress the scale and stabilise variance.

### 4.6.3 Effect of Transformations on Shape

- **Square-root:** compresses large values moderately, shortens right tail  $\Rightarrow$  reduces mild right-skewness.
- **Logarithm:** compresses large values more strongly, effective for heavily right-skewed data or data spanning many orders of magnitude.
- **Square:** stretches upper tail  $\Rightarrow$  reduces left-skewness (but may introduce some right-skewness).

### Diagnosing Non-Linearity: Always Check Residuals

- The scatterplot may appear roughly linear while the true relationship is not.
- The **residual plot** ( $\hat{y}_i$  vs.  $e_i$ ) is more sensitive: a curved pattern signals non-linearity even when the scatterplot looks acceptable.
- *Always examine the residual plot after fitting a model.*

### Cautions When Transforming

- Transformations are **not a cure-all**. They may not fully resolve non-linearity or heteroscedasticity; in severe cases, a genuinely non-linear model or weighted least squares may be required.
- Transformed variables are harder to interpret for non-technical audiences.
- There is rarely a single “correct” transformation — experiment with alternatives and balance model fit against interpretability.
- **Critical:** all estimates and confidence intervals are on the *transformed* scale. Convert back to the original scale before reporting results.

## Summary Table

| Topic                        | Key Points                                                                                                                     |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Polynomial regression</b> | Still MLR in disguise; treat $x^2, x^3, \dots$ as new predictors.                                                              |
| <b>Hierarchy</b>             | Model must include all lower-order terms up to the highest order used.                                                         |
| <b>VIF</b>                   | $VIF_j = 1/(1 - R_j^2)$ ; $> 5$ moderate, $> 10$ severe multicollinearity.                                                     |
| <b>Model selection</b>       | Compare $R_{adj}^2$ , $t$ -tests on highest term, VIF; prefer parsimony.                                                       |
| <b>Linearisation</b>         | Log, reciprocal, square root transforms can convert non-linear models to linear ones.                                          |
| <b>Semi-log</b>              | $\log Y = \beta_0 + \beta_1 X$ : 1-unit $\uparrow$ in $X$ multiplies $Y$ by $e^{\beta_1}$ .                                    |
| <b>Log-log</b>               | $\log Y = \beta_0 + \beta_1 \log X$ : 1% $\uparrow$ in $X$ gives $\beta_1\%$ $\uparrow$ in $Y$ (elasticity).                   |
| <b>Skewness</b>              | Right-skewed $\Rightarrow \alpha < 1$ (log, $\sqrt{\cdot}$ , reciprocal); left-skewed $\Rightarrow \alpha > 1$ ( $y^2, y^3$ ). |
| <b>Extrapolation</b>         | Polynomial curvature outside data range is unreliable — avoid.                                                                 |
| <b>Scale conversion</b>      | Always back-transform estimates and CIs to the original scale for interpretation.                                              |